

Skills Summary

Transformations of the Absolute-Value Graph

Use this reference while completing your worksheet or when studying.

Skill 1 — Moving the V left, right, up and down

Parent graph: $y = |x|$ is a V with its corner at the origin and arms of slope 1 and -1 .

Form: $y = |x - h| + k$ puts the corner at (h, k) . The corner is wherever the *inside of the bars is zero*, which is why $|x - 3|$ sits three units to the right and $|x + 3|$ three units to the left.

Example: Find the x -intercepts of $y = |x - 2| - 3$.

Step 1. An intercept means $y = 0$, so isolate the bars first — and expect two answers, because two numbers lie the same distance from zero.

Step 2. $|x - 2| = 3$ splits into $x - 2 = 3$ and $x - 2 = -3$, giving $x = 5$ and $x = -1$.

$$x = -1 \text{ or } x = 5$$

Try it: Find the x -intercepts of $y = |x + 3| - 2$.

check: $1 - = x$ to $2 - = x$

(!) **Watch out:** the corner of $y = |x + 3| - 2$ is at $x = -3$, not $x = 3$. Ask “what makes the inside zero?” instead of reading the sign, and the direction is never wrong.

Skill 2 — Flipping and steepening the arms

Form: $y = a|x - h| + k$. The number a acts on the arms only, never on the corner's position.

$a < 0$ turns the V upside down. $|a| > 1$ makes the arms steeper (narrower V); $|a| < 1$ makes them shallower (wider V). Substituting: bars first, then multiply by a , then add k .

Example: For $y = -2|x| + 6$, find the value at $x = 1$.

Step 1. This is a substitution, so the only decision is order — the bars close before the -2 is applied, so take the distance first.

Step 2. $|1| = 1$; then $-2 \cdot 1 = -2$; then $-2 + 6$.

$$y = 4$$

Try it: For $y = 3|x - 1| - 5$, find the value at $x = 3$.

check: $1 = 6$

Skill 3 — Drawing the graph from the equation

Three steps, always in this order: plot the corner (h, k) ; find the x -intercepts by solving $y = 0$; join each intercept to the corner with a *straight* arm and continue it.

The two intercepts are always the same distance from the corner — if yours are not, a sign is wrong.

Example: Where does $y = -|x - 1| + 4$ cross the horizontal axis?

Step 1. The corner is at $(1, 4)$ and the minus flips the V downward, so the graph must come back down to the axis on both sides — set $y = 0$ to find where.

Step 2. $|x - 1| = 4$ gives $x - 1 = \pm 4$, so $x = 5$ and $x = -3$, each four units from the corner.
 $x = -3$ or $x = 5$

Try it: Where does $y = |x + 1| - 3$ cross the horizontal axis?

Check: $x = -4$ or $x = 2$